

# Thermo Scientific™ HyPerforma™ Rocker with 20 L BPC - 2-10 L working volume

thermo-rocker-20

## Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

| Parameter                                   | Value in simulator                  | Evidence status     |
|---------------------------------------------|-------------------------------------|---------------------|
| Minimum working volume                      | 2 L                                 | Source-supported    |
| Maximum working volume                      | 10 L                                | Source-supported    |
| Nominal vessel / bag size                   | 20 L                                | Source-supported    |
| Total vessel volume loaded into the form    | 20 L                                | Source-supported    |
| Initial working volume loaded into the form | 2 L                                 | Derived / estimated |
| Product-contact material                    | Plastic / single-use                | App assumption      |
| Height : diameter                           | 0.7                                 | App assumption      |
| Internal / chamber diameter                 | 0.263 m                             | Derived / estimated |
| Liquid height at maximum working volume     | 0.1841 m                            | Derived / estimated |
| Impeller / mixing type                      | Marine / placeholder mixing element | App assumption      |
| Impeller count                              | 1                                   | App assumption      |
| Impeller diameter                           | 0.0868 m                            | Derived / estimated |
| Power number                                | 0.15                                | App assumption      |
| Initial agitation                           | 20 rpm                              | App assumption      |
| Maximum agitation                           | 40 rpm                              | App assumption      |
| Initial aeration                            | 0.05 vvm                            | App assumption      |
| Maximum aeration                            | 0.5 vvm                             | App assumption      |
| Baffled vessel                              | No                                  | App assumption      |
| Maximum pressure                            | 0.1 bar(g)                          | App assumption      |
| Default operating pressure                  | 0 bar(g)                            | App assumption      |
| Maximum inlet oxygen fraction               | 100%                                | App assumption      |

## Parameter evidence and derivations

### Minimum working volume - 2 L

Source-supported | Thermo HyPerforma Rocker datasheet - PDF page 2; 20 L BPC column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 2 L.

Limitations: Direct tabulated minimum and working volume. Rocking rate 2-40 cycles/min and angle 2-12 degrees per side are not impeller specifications.

Local original: [Thermo HyPerforma Rocker datasheet - PDF page 2](#)

[Manufacturer publication](#)

### Maximum working volume - 10 L

Source-supported | Thermo HyPerforma Rocker datasheet - PDF page 2; 20 L BPC column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: Direct tabulated minimum and working volume. Rocking rate 2-40 cycles/min and angle 2-12 degrees per side are not impeller specifications.

Local original: [Thermo HyPerforma Rocker datasheet - PDF page 2](#)

[Manufacturer publication](#)

### Nominal vessel / bag size - 20 L

Source-supported | Thermo HyPerforma Rocker datasheet - PDF page 2; 20 L BPC column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 20 L.

Limitations: Direct tabulated minimum and working volume. Rocking rate 2-40 cycles/min and angle 2-12 degrees per side are not impeller specifications.

Local original: [Thermo HyPerforma Rocker datasheet - PDF page 2](#)

[Manufacturer publication](#)

### Total vessel volume loaded into the form - 20 L

Source-supported | Thermo HyPerforma Rocker datasheet - PDF page 2; 20 L BPC column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 20 L.

Limitations: Direct tabulated minimum and working volume. Rocking rate 2-40 cycles/min and angle 2-12 degrees per side are not impeller specifications.

Local original: [Thermo HyPerforma Rocker datasheet - PDF page 2](#)

[Manufacturer publication](#)

### Initial working volume loaded into the form - 2 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation:  $\min(2 \text{ L}, 10 \text{ L}) = 2 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

### Product-contact material - Plastic / single-use

App assumption | Unverified legacy preset

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Plastic / single-use.

Limitations: No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

## Height : diameter - 0.7

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].heightDiameterRatio = 0.7

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This ratio is subsequently used to estimate vessel diameter and liquid height. This is a rocking-bag system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

## Internal / chamber diameter - 0.263 m

Derived / estimated | Cylindrical geometry estimate

Rule:  $D = \sqrt[3]{(4 \times V) / (\pi \times H:D)}$

Selection / calculation:  $D = \sqrt[3]{(4 \times 0.01 \text{ m}^3) / (\pi \times 0.7)} = 0.2629889 \text{ m}$ ; the form displays 0.263 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

## Liquid height at maximum working volume - 0.1841 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation:  $0.2629889 \text{ m} \times 0.7 = 0.1840922 \text{ m}$ ; the form displays 0.1841 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

## Impeller / mixing type - Marine / placeholder mixing element

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].impellerType = "marine"

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. The selected type determines the generic mixing archetype shown and is paired with the preset power-number assumption. This is a rocking-bag system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

## Impeller count - 1

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].impellerCount = 1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported. This is a rocking-bag system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

## Impeller diameter - 0.0868 m

Derived / estimated | Generic impeller-to-tank ratio

Rule: impeller diameter =  $\max(0.33 \times \text{estimated vessel diameter}, 0.001 \text{ m})$

Selection / calculation:  $\max(0.33 \times 0.2629889 \text{ m}, 0.001 \text{ m}) = 0.08678633 \text{ m}$ ; the form displays 0.0868 m.

Limitations: A one-third tank-diameter impeller is used only as a generic starting geometry when no model-specific impeller diameter is stored.

## Power number - 0.15

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].powerNumber = 0.15

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. The power number is an empirical generic value used in  $P = N_p N_3 D_i^5$ . It is not recalculated from blade geometry. This is a rocking-bag system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

## Initial agitation - 20 rpm

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].baseRpm = 20

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This is the initial agitation setting loaded into the form. This is a rocking-bag system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

## Maximum agitation - 40 rpm

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].maxRpm = 40

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This is the ceiling available to the generic DO-control cascade. This is a rocking-bag system. These stirred-tank fields are compatibility placeholders for the current simulation engine and should not be interpreted as physical impeller specifications.

## Initial aeration - 0.05 vvm

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].baseVvm = 0.05

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This is the initial gas rate loaded into the form.

## Maximum aeration - 0.5 vvm

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].maxVvm = 0.5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

## Baffled vessel - No

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].baffled = false

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

## Maximum pressure - 0.1 bar(g)

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].maxPressure = 0.1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

## Default operating pressure - 0 bar(g)

App assumption | Generic rocking-bag profile

Rule: PROFILE\_DEFAULTS["rocking"].operatingPressure = 0

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable rocking-bag starting profile. This is the starting headspace pressure loaded into the form.

## Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

## Configuration notes and unresolved evidence

This is a rocking system with no conventional impeller; impeller fields are placeholders required by the current stirred-tank engine.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

PDF p. 3 distinguishes LDPE sensor BPCs and Aegis5-14 sensorless BPCs. The preset does not choose a film configuration.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

## Local original document identities

Thermo HyPerforma Rocker datasheet

thermo-hyperforma-rocker-datasheet-fab5ad31faf1.pdf

SHA-256: fab5ad31faf178cdb7c6a34562c0d2bbfb7ca9d60177d799a993212d02ffaf3c